Name ______________________________________________ Date ___________ Period _____

# Domain and Range: All Three Notations

Algebra II | Unit 1 | Day 1 | TEK A2.7.I

**Objective:** I can write the domain and range of a function from a graph, a table, or a set of ordered pairs using a verbal description, inequality notation, set notation, and interval notation.

## Vocabulary

**Domain** — the set of all _______________ values, the _____-values.

**Range** — the set of all _______________ values, the _____-values.

**Set notation** — uses ___________________ with the bar | , read "such that."

**Interval notation** — uses _________ if an endpoint is included, _________ if it is excluded.

**Inequality notation** — writes domain or range using _____________________ instead of interval or set notation.

## Part 1 — From a Set of Ordered Pairs

*Least to greatest, and each value written only once.*

**Example 1.** { (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) }

Domain: { ____________________________________ }

Range: { ____________________________________ }

**You Try 1.** { (-5, 3), (-1, 0), (2, 3), (6, -8) }

Domain: { ____________________________________ }

Range: { ____________________________________ }

## Part 2 — From a Table

**Example 2.**

| x | -3 | -1 | 0 | 2 | 5 |
|---|---|---|---|---|---|
| f(x) | 9 | 1 | 0 | 4 | 25 |

Domain: { ____________________________________ } Range: { ________________________________ }

**You Try 2.**

| x | -2 | 0 | 1 | 4 |
|---|---|---|---|---|
| f(x) | 7 | 7 | -3 | 10 |

Domain: { ____________________________________ } Range: { ________________________________ }

## Part 3 — The Three Notations

*The first row is done for you. Fill in the rest.*

| Verbal description | Inequality | Set notation | Interval notation |
|---|---|---|---|
| all real numbers | $-\infty < x < \infty$ | $\{ x \mid x \in \mathbb{R} \}$ | $(-\infty, \infty)$ |
| all reals greater than or equal to 0 | $x \geq 0$ | ____________________ | ____________ |
| all reals less than 5 | $x < 5$ | ____________________ | ____________ |

| | | | |
|---|---|---|---|
| all reals from -3 to 3, including both | $-3 \le x \le 3$ | ____________________ | ____________ |
| just the numbers -2, 0, and 4 | *(none)* | ____________________ | ____________ |

**Key idea:** bracket [ ] = endpoint _______________. Parenthesis ( ) = endpoint _______________. ∞ always takes a _______________.

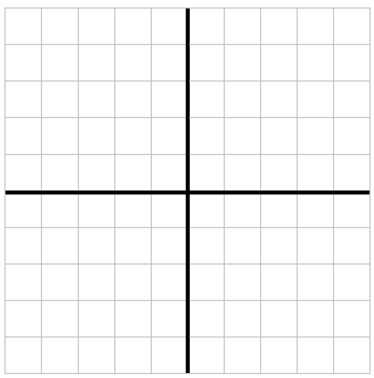
**You Try 3.** all reals greater than -8 → inequality _________________ interval _________________

**You Try 4.** $y \le 12$ → in words ______________________________ interval _______________

## Part 4 — From a Graph

*Each square is 1 unit. The axes are the two heavy lines, and both grids run -5 to 5.*

**A. Discrete.** Plot (-4, 2), (-2, -3), (1, 4), (3, -1).

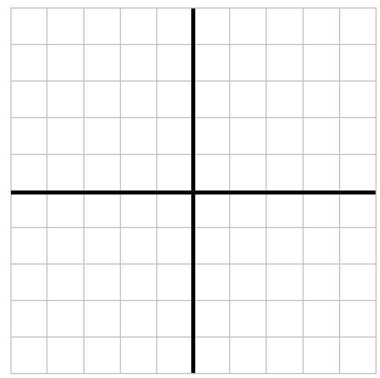


Domain (set): _________________

Range (set): _________________

Continuous or discrete? _______________

**B. Continuous.** Graph $y = x + 1$ from $x = -4$ to $x = 3$.



Domain (inequality): _________________

Domain (interval): _________________

Range (interval): _________________

## Part 5 — Continuous or Discrete?

*Write continuous or discrete.*

1. The number of students in each class period ______________________
2. The outside temperature over one full day ______________________
3. The total cost of n concert tickets at $45 each ______________________

## Exit Ticket

Given { (1, 5), (3, 5), (4, -2) } write the domain and range in set notation, then say whether interval notation could be used.

Domain ________________________ Range ________________________

Could interval notation be used? _________________________________

Name ______________________________________________ Date ___________ Period _____

# Domain and Range from Problem Situations

Algebra II | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

**Objective:** I can find the domain and range of a real-world situation, explain what they mean in context, and tell the difference between the domain of a function and the domain that the situation allows.

## Part 1 — Name the Variables

*Which quantity is the input and which is the output?*

1. Hours you work and the money you earn
   independent: _______________________ dependent: _______________________

2. Number of tickets bought and the total cost
   independent: _______________________ dependent: _______________________

3. Seconds since a ball was thrown and its height
   independent: _______________________ dependent: _______________________

## Part 2 — Function Domain vs. Contextual Domain

A cube-shaped box has side length x inches. Its volume is **$V(x) = x^3$**.

What domain does the **equation** $x^3$ accept? _________________________________

What domain does the **box** allow? _________________________________

Why? ________________________________________________

**Big idea:** the equation may accept numbers the _______________ does not. Always ask what makes sense.

## Part 3 — Domain and Range in Context

*Give the contextual domain and range. Use an inequality and interval notation.*

**Example 1.** Concert tickets cost \$12 each. You may buy at most 8. Total cost $C(t) = 12t$.

Domain: ____________________________________________ Discrete or continuous? _______________

Range: ______________________________________________

**You Try 1.** A ball is thrown. Its height is $h(t) = -16t^2 + 64t$ feet, and it is in the air from $t = 0$ to $t = 4$ seconds. Its greatest height is 64 feet.

Domain: ___________________ interval _______________ Discrete or continuous? _______________

Range: ___________________ interval _______________

**You Try 2.** A parking garage charges \$3 per hour and is open 6 a.m. to 10 p.m. (16 hours).

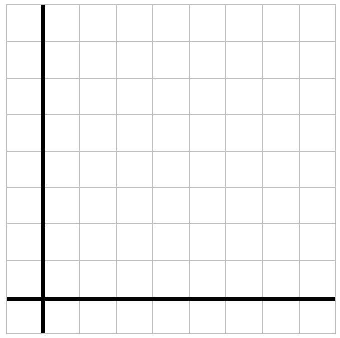
Domain: ____________________________ Range: ____________________________

## Part 4 — Reading a Situation from a Graph

*Each square is 1 unit. Plot the points, then answer in context.*

**A water tank is drained.** Gallons g after m minutes:

| **m** | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| **g** | 8 | 6 | 4 | 2 | 0 |



**Answer in context.**

Domain: ____________________

Range: ____________________

What does the domain mean here?

______________________________

What does a range of 0 mean?

______________________________

## Part 5 — Does the Situation Restrict the Domain?

*Write yes or no, and if yes, give the restriction.*

1. The side length of a square garden ____________________________
2. The temperature in a freezer over one day ____________________________
3. The number of buses needed for a field trip ____________________________

## Practice

*Give the contextual domain and range for each. Show your thinking on the lines.*

**1.** A phone plan costs $30 per month plus $5 per gigabyte, up to 20 GB.

______________________________________________________________

Domain ____________________________ Range ____________________________

**2.** A theater has 250 seats. Tickets are $9 each, and t tickets are sold.

______________________________________________________________

______________________________________________________________

Domain ____________________________ Range ____________________________

**3.** A diver descends from the surface to 40 meters below sea level.

______________________________________________________________

______________________________________________________________

Range of depth ____________________________ Why negative? _____________________

## Exit Ticket

A taxi charges a $4 flat fee plus $2 per mile, for trips up to 10 miles. Cost $C(m) = 4 + 2m$.

Domain ____________________________ Range ____________________________

Why does the range start at 4, not 0? ____________________________________

Name ______________________________________________ Date ___________ Period _____

# Continuous vs. Discrete

Algebra II | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

**Objective:** I can decide whether a domain and range are continuous or discrete, and write each one using the notation that fits.

## Vocabulary

**Continuous function** — the graph is _________________ — you could trace it without lifting your pencil.

**Discrete function** — the graph is made of separate, ___________________ points.

## Part 1 — Sort the Situations

*Write C for continuous or D for discrete.*

| Situation | C or D | Why? |
|---|---|---|
| The number of chairs in a classroom | _____ | ____________________ |
| The distance a car travels | _____ | ____________________ |
| Your age measured in years and days | _____ | ____________________ |
| The number of pets a family owns | _____ | ____________________ |
| The weight of a watermelon | _____ | ____________________ |

## Part 2 — The Same Function, Two Ways

A cube-shaped box has side x inches, so its volume is **$V(x) = x^3$**.

| x | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $V(x) = x^3$ | 1 | 8 | 27 | 64 |

**Case A:** the box can be any size at all.
Domain __________________________ Range __________________________ _______________

**Case B:** the box is built out of 1-inch cube blocks, so x must be a whole number.
Domain __________________________ Range __________________________ _______________

Same equation, different _______________ → different domain. The context decides.

## Part 3 — Which Notation Fits?

Continuous domains use _______________ notation, like [0, 5].

Discrete domains get _______________ in braces, like { 0, 1, 2, 3 }, because there are gaps between the values.

**You Try 1.** A stadium sells 0 to 5 hot dogs per person. Domain __________________________

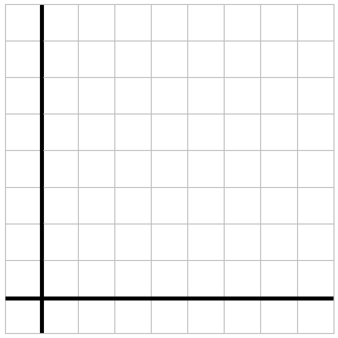
**You Try 2.** A candle burns from 10 cm down to 0 cm. Domain of heights _____________________

**You Try 3.** Why can You Try 1 not use interval notation? ________________________________

## Part 4 — Graph Both Kinds

*Each square is 1 unit. Plot each, then name the domain and range.*

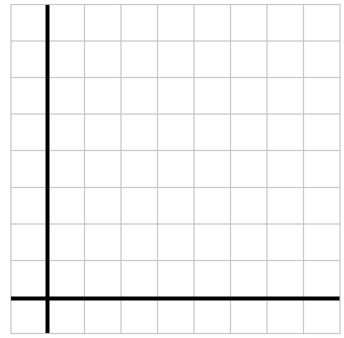
**A. Discrete.** A tutor charges $2 per session, up to 4 sessions. Plot (1, 2), (2, 4), (3, 6), (4, 8).



Domain ___________________

Range ___________________

**B. Continuous.** A candle burns 2 cm per hour from 8 cm. Graph $h = 8 - 2t$ for $0 \le t \le 4$.



Domain ___________________

Range ___________________

## Part 5 — Quick Check

1. Should a discrete domain ever be written as an interval? ____________________________
2. Can a continuous range include negative numbers? ____________________________
3. Is money spent at a vending machine continuous? ____________________________

## Practice

*Decide continuous or discrete, then write the domain in the notation that fits.*

**1.** A runner covers any distance from 0 to 5 miles.

C or D ________ Domain __________________________ Why? ____________________________

**2.** A vending machine sells 0 to 6 drinks.

C or D ________ Domain __________________________ Why? ____________________________

**3.** Water in a bathtub drops from 30 gallons to 0.

C or D ________ Range __________________________ Why? ____________________________

**4.** A class of 24 students is split into equal teams of 4.

C or D ________ Domain (teams) __________________________

**5.** Explain in your own words why a discrete domain cannot use interval notation.

_______________________________________________________________

_______________________________________________________________

## Exit Ticket

A school orders pizzas for a party. Each pizza feeds 6 students, and they order up to 5 pizzas.

Domain (pizzas) __________________________ Range (students fed) ____________________________

Continuous or discrete, and why? ____________________________________

Name ______________________________________________ Date ___________ Period _____

# Parent Functions I: $\sqrt{x}$ and $|x|$

Algebra II | Unit 1 | Day 4 | TEK A2.2.A

**Objective:** I can graph the square root and absolute value parent functions from a table and analyze their key attributes: domain, range, intercepts, symmetry, and minimum.

## Vocabulary

**Reflectional symmetry** — the graph can be _____________ across a line and land exactly on itself.

**x-intercept(s)** — where the graph crosses the x-axis. There, y = _____. **y-intercept(s)** — there, x = _____.

**Zeros** — the x-value(s) that make a function equal _____ — the same locations as the x-intercepts.

## Part 1 — The Square Root Function, $f(x) = \sqrt{x}$

*Fill in the table. Watch what happens with negative inputs.*

| **x** | -4 | -1 | 0 | 1 | 4 | 9 |
|---|---|---|---|---|---|---|
| **$f(x) = \sqrt{x}$** | | | | | | |

Why are the first two undefined? _____________________________________________________

## Part 2 — The Absolute Value Function, $f(x) = |x|$

*Absolute value gives distance from zero, so the output is never negative.*

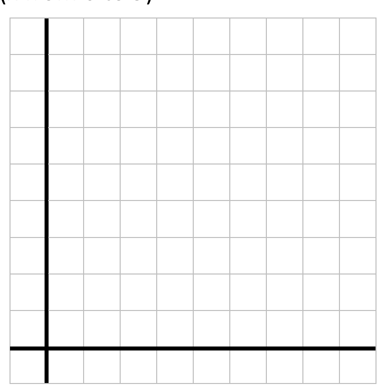
| **x** | -5 | -2 | -1 | 0 | 1 | 2 | 5 |
|---|---|---|---|---|---|---|---|
| **$f(x) = \lvert x \rvert$** | | | | | | | |

What shape does this make? ____________________________________________

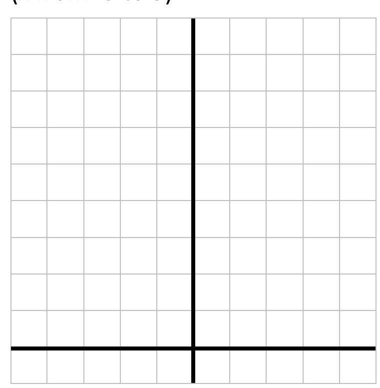
## Part 3 — Graph Them Both

*Each square is 1 unit. Plot your table values, then connect them smoothly.*

**A. $f(x) = \sqrt{x}$** (x from 0 to 9)



**B. $f(x) = |x|$** (x from -5 to 5)



Which quadrant only? ________

Fold line? ______________

## Part 4 — Compare the Key Attributes

| **Attribute** | **$f(x) = \sqrt{x}$** | **$f(x) = \lvert x \rvert$** |
|---|---|---|
| Domain (interval) | ________________ | ________________ |

| | | |
|---|---|---|
| Range (interval) | ________________ | ________________ |
| x- and y-intercept | ________________ | ________________ |
| Symmetry | ________________ | ________________ |
| Minimum | ________________ | ________________ |
| Increasing / decreasing | ________________ | ________________ |

**Biggest difference:** their _______________. Only ___________ refuses negative inputs.

## Part 5 — Name That Function

*Use only the attributes given. No equation.*

1. Domain and range both [0, ∞), no symmetry, always increasing. ____________________
2. Domain all reals, range [0, ∞), symmetric across the y-axis, minimum 0. ____________________

## Practice

*Evaluate, then answer. Show your work where there is room.*

**1.** Evaluate: √25 = _____________ √0 = _____________ |-7| = _____________ |0| = _____________

**2.** Explain why √-9 is not a real number.

_________________________________________________________________

_________________________________________________________________

**3.** Name every x-value where |x| = 4. _____________________________ How many? _________

**4.** Name every x-value where √x = 4. _____________________________ How many? _________

**5.** Items 3 and 4 have different numbers of answers. What does that tell you about the two graphs?

_________________________________________________________________

_________________________________________________________________

**6.** On [1, 4], give the minimum and maximum of f(x) = √x. min _____________ max _____________

## Exit Ticket

Both functions have a minimum of 0 and pass through the origin. Give one attribute that tells them apart, and explain why it happens.

Attribute: _________________________________

Why: ___________________________________________________

Name ______________________________________________ Date ___________ Period _____

# Parent Functions II: $x^3$ and $\sqrt[3]{x}$

Algebra II | Unit 1 | Day 5 | TEK A2.2.A

**Objective:** I can graph the cubic and cube root parent functions from a table and analyze their key attributes, including their rotational symmetry about the origin.

## Part 1 — The Cubic Function, $f(x) = x^3$

*Fill in the table. Notice what a negative input produces.*

| **x** | -2 | -1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| **$f(x) = x^3$** | | | | | |

A negative input gives a _______________ output, because a negative cubed stays negative.

## Part 2 — The Cube Root Function, $f(x) = \sqrt[3]{x}$

*This undoes the cubic. Compare it to your Part 1 table.*

| **x** | -8 | -1 | 0 | 1 | 8 |
|---|---|---|---|---|---|
| **$f(x) = \sqrt[3]{x}$** | | | | | |

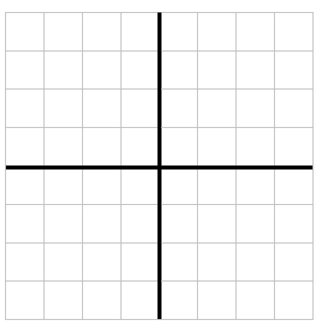
What do you notice next to Part 1? ___________________________________________

We will come back to that on Day 10 - these two are _______________.

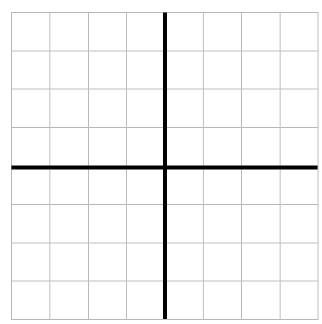
## Part 3 — Graph Them Both

*Each square is 1 unit. Both grids run -8 to 8 on each axis, so 1 square = 2 units.*

**A. $f(x) = x^3$** steep S-curve



**B. $f(x) = \sqrt[3]{x}$** flatter S-curve



Passes through the _____________

Mirror line between them: _____________

## Part 4 — Compare the Key Attributes

| **Attribute** | **$f(x) = x^3$** | **$f(x) = \sqrt[3]{x}$** |
|---|---|---|
| Domain (interval) | ________________ | ________________ |
| Range (interval) | ________________ | ________________ |
| x- and y-intercept | ________________ | ________________ |
| Symmetry | ________________ | ________________ |
| Maximum or minimum? | ________________ | ________________ |

| | | |
|---|---|---|
| Increasing / decreasing | ________________ | ________________ |

These two share every attribute above. The difference is the ______________ of the curve.

## Part 5 — Compare to Day 4

1. Why does $\sqrt[3]{x}$ accept negatives when $\sqrt{x}$ does not? ____________________________________________
2. Which of the four functions so far have a minimum? __________________________

## Practice

*Evaluate and explain. Show your work where there is room.*

**1.** Evaluate: $(-3)^3$ = _____________ $\sqrt[3]{-27}$ = _____________ $\sqrt[3]{64}$ = _____________ $\sqrt[3]{0}$ = _____________

**2.** Solve $x^3 = 125$. x = _______________ How many real answers? ________

**3.** Solve $x^2 = 25$. x = _______________ How many real answers? ________

**4.** Items 2 and 3 give different counts. Explain why cubing behaves differently from squaring.

_______________________________________________________________

_______________________________________________________________

**5.** $f(x) = x^3$ on the interval $[-1, 3]$: minimum _____________ maximum _____________

**6.** True or false: every real number has exactly one real cube root. _______________

**7.** Which of the four parent functions so far accept negative inputs?

_____________________________________________

**8.** Solve $x^3 = -8$. x = _____________ How many real answers? ________

**9.** Evaluate: $\sqrt[3]{-1000}$ = _____________

**10.** A cube-shaped storage box has volume $V = x^3$. If $V = 343$ in$^3$, what is the side length x? _______________

**11.** True or false: a negative number has exactly one real cube root, but no real square root. _______________

## Exit Ticket

A function has domain and range both $(-\infty, \infty)$, passes through the origin, has 180° rotational symmetry, and no maximum or minimum.

Name two parent functions this could be: ____________________________

What extra information would tell them apart? _______________________________________

Name ______________________________________________ Date ___________ Period _____

# Parent Functions III: 1/x and Asymptotes

Algebra II | Unit 1 | Day 6 | TEK A2.2.A

**Objective:** I can graph the reciprocal parent function, identify its vertical and horizontal asymptotes, and describe its domain, range, and symmetry.

## Vocabulary

**Asymptotic behavior** — how a graph behaves near a value it gets closer and closer to but never _______________.

**Vertical asymptote** — a vertical line the graph _________________. It sits where the function is undefined.

**Horizontal asymptote** — a horizontal line the y-values get ________________________________, as x gets very large or very small.

## Part 1 — Build the Table for f(x) = 1/x

*Fill in the outputs. One input has no output at all.*

| x | -2 | -1 | -1/2 | 0 | 1/2 | 1 | 2 |
|---|---|---|---|---|---|---|---|
| **f(x) = 1/x** | | | | | | | |

Why is x = 0 undefined? ___________________________________________

## Part 2 — What Happens Near Zero and Far Out

*Follow the pattern in each list, then describe it.*

| x | 1 | 0.5 | 0.1 | 0.01 |
|---|---|---|---|---|
| **1/x** | 1 | 2 | 10 | 100 |

As x gets close to 0, the outputs grow ___________________. So there is a vertical asymptote at __________.

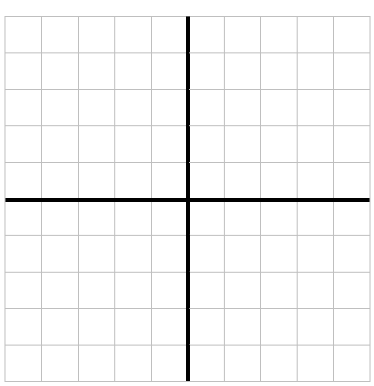
| x | 1 | 10 | 100 | 1000 |
|---|---|---|---|---|
| **1/x** | 1 | 0.1 | 0.01 | 0.001 |

As x gets very large, the outputs approach ________. So there is a horizontal asymptote at ___________.

## Part 3 — Graph It

*Each square is 1 unit. Plot your table points. You will get two separate branches - do not connect them across the y-axis.*

**f(x) = 1/x**



**Record what you see.**

Vertical asymptote: ______________

Horizontal asymptote: ______________

How many branches? ________

Which quadrants? ______________

Any intercepts? ______________

*Draw both asymptotes as dashed lines.*

## Part 4 — Key Attributes of f(x) = 1/x

Domain: ___________________________ set notation ___________________________

Range: ___________________________ interval ___________________________

Symmetry: reflectional across _______________________, and rotational ________ about the origin.

Maximum or minimum? ___________________ Intercepts? ___________________

This is the first parent function with ___________________ and the first whose domain has a __________ in it.

## Part 5 — True or False

1. A graph may cross a vertical asymptote. _________________
2. A graph may cross a horizontal asymptote. _________________
3. f(x) = 1/x has a y-intercept. _________________
4. The domain of 1/x is every real number. _________________

## Practice

*Evaluate and explain. Show your work where there is room.*

**1.** Evaluate: 1/4 = _____________ 1/(-2) = _____________ 1/(1/5) = _____________

**2.** Solve 1/x = 4. x = _________________ Solve 1/x = -1. x = _________________

**3.** Is there any x with 1/x = 0? Explain.

______________________________________________________________

______________________________________________________________

**4.** As x grows very large and negative, what happens to 1/x? ________________________________

**5.** Fill in the pattern.

| **x** | -0.1 | -0.01 | 0.01 | 0.1 |
|---|---|---|---|---|
| **1/x** | | | | |

Approaching 0 from the left the outputs go toward _____________, from the right toward _____________.

**6.** Why is 1/x the only parent function so far with a gap in its domain?

______________________________________________________________

______________________________________________________________

## Exit Ticket

A function has no intercepts, a vertical asymptote at x = 0, a horizontal asymptote at y = 0, and two branches in Quadrants I and III.

Name it: __________________________ Domain in set notation: ____________________________

Name ______________________________________________ Date ___________ Period _____

# Parent Functions IV: $b^x$ and log base b

Algebra II | Unit 1 | Day 7 | TEK A2.2.A

**Objective:** I can graph exponential and logarithmic parent functions for b = 2, 10, and e, analyze their key attributes, and notice that their domains and ranges trade places.

## Part 1 — The Exponential Function, $f(x) = 2^x$

*Fill in the table. Negative exponents give fractions, never negative outputs.*

| **x** | -2 | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|
| **$f(x) = 2^x$** | | | | | | |

Can $2^x$ ever equal 0 or a negative number? ____________________________________________

So there is a horizontal asymptote at _____________ and a y-intercept at _____________.

## Part 2 — The Logarithmic Function, $f(x) = \log_2(x)$

*This undoes the exponential. Compare it to your Part 1 table.*

| **x** | 1/4 | 1/2 | 1 | 2 | 4 | 8 |
|---|---|---|---|---|---|---|
| **$f(x) = \log_2(x)$** | | | | | | |

What happened to the table from Part 1? ____________________________________________

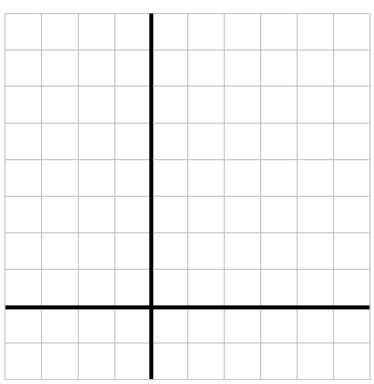
What is $\log_2(0)$? __________________________________________

So there is a vertical asymptote at _____________ and an x-intercept at _____________.

## Part 3 — Graph Them Both

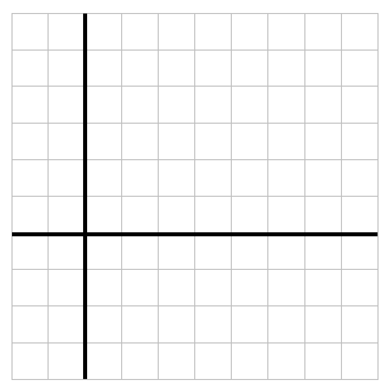
*Each square is 1 unit. Sketch the asymptote as a dashed line first, then plot.*

**A. $f(x) = 2^x$**



Asymptote: _____________

**B. $f(x) = \log_2(x)$**



Asymptote: _____________

## Part 4 — Compare the Key Attributes

| **Attribute** | **$f(x) = b^x$** | **f(x) = log base b** |
|---|---|---|
| Domain (interval) | ________________ | ________________ |

| | | |
|---|---|---|
| Range (interval) | ________________ | ________________ |
| Intercept | ________________ | ________________ |
| Asymptote | ________________ | ________________ |
| Increasing / decreasing | ________________ | ________________ |

Look at the first two rows. The domain and range have ________________, and so did the intercepts and asymptotes. That is a strong hint these are ____________ - Day 10.

## Part 5 — Changing the Base

*All bases greater than 1 give the same shape. Only the steepness changes.*

| **x** | 0 | 1 | 2 |
|---|---|---|---|
| **$2^x$** | 1 | 2 | 4 |

| **x** | 0 | 1 | 2 |
|---|---|---|---|
| **$10^x$** | 1 | 10 | 100 |

Which grows faster, and why? _____________________________________________

What do all three exponential graphs share? _______________________________

## Practice

*Show your work where there is room.*

**1.** From your Part 1 table, a negative exponent on 2 always gives a _______________________________, never a negative number.

**2.** Evaluate: $\log_2(8)$ = ____________ (because $2^3 = 8$)

**3.** Solve $2^x = 16$. x = ____________

**4.** Solve $\log_2(x) = 5$. x = ____________

**5.** A function has domain $(0, \infty)$, range all reals, and passes through (1, 0). Name it. _______________________

**6.** A function has a horizontal asymptote y = 0 and y-intercept (0, 1). Name it. _______________________

**7.** True or false: $\log_2(0)$ is defined. ______________

**8.** Explain why $\log_2(0)$ is undefined - what power of 2 could possibly give 0?

____________________________________________________________

____________________________________________________________

## Exit Ticket

A function has domain $(0, \infty)$, range $(-\infty, \infty)$, an x-intercept at (1, 0), and a vertical asymptote at x = 0.

Name the family: _________________________ Name its inverse family: _________________________

Name ____________________________________________ Date ___________ Period _____

# Comparing All Seven Parent Functions

Algebra II | Unit 1 | Day 8 | TEK A2.2.A

**Objective:** I can compare the key attributes of all seven parent functions, name a function from its attributes alone, and find a relative maximum or minimum on a given interval.

## Part 1 — The Master Comparison Table

*Fill this in from your Day 4-7 notes. This is your reference sheet for the rest of the year.*

| Function | Domain | Range | Symmetry | Asymptotes |
|---|---|---|---|---|
| **√x** | __________ | __________ | __________ | __________ |
| **\|x\|** | __________ | __________ | __________ | __________ |
| **$x^3$** | __________ | __________ | __________ | __________ |
| **∛x** | __________ | __________ | __________ | __________ |
| **1/x** | __________ | __________ | __________ | __________ |
| **$b^x$** | __________ | __________ | __________ | __________ |
| **log base b** | __________ | __________ | __________ | __________ |

## Part 2 — Intercepts and Extremes

| Function | Intercept(s) | Minimum? | Maximum? |
|---|---|---|---|
| √x | ____________ | ____________ | __________ |
| \|x\| | ____________ | ____________ | __________ |
| $x^3$ and ∛x | ____________ | ____________ | __________ |
| 1/x | ____________ | ____________ | __________ |
| $b^x$ | ____________ | ____________ | __________ |
| log base b | ____________ | ____________ | __________ |

Only ________ of the seven have a minimum, and ________ has a maximum. Only ________ has no intercept at all.

## Part 3 — Name That Function

*Use only the attributes listed. Name the function and sketch its shape in the box.*

1. Domain and range both all reals except 0; two branches; asymptotes at x = 0 and y = 0.
   Function: ______________________

2. Domain all reals, range [0, ∞), folds across the y-axis, minimum of 0.
   Function: ______________________

3. Domain (0, ∞), range all reals, passes through (1, 0).
   Function: ______________________

4. Domain and range both [0, ∞), no symmetry, no asymptotes.
   Function: ______________________

## Part 4 — Relative Max and Min on an Interval

*A graph rises and falls between y = 1.5 and y = -1.5 over the interval from 0° to 360°.*

Relative maximum on that interval: ___________________ Relative minimum: ___________________

Now consider f(x) = $x^3$ on the interval [-2, 2].

Relative maximum: ______________________ Relative minimum: ______________________

$x^3$ has no overall max or min, but on a __________________ it has both. Why? ________________________

## Part 5 — Sort by One Attribute

Which parent functions have a restricted domain? ____________________________

Which have any asymptote? ____________________________

Which have range all real numbers? ____________________________

## Practice

*Use your comparison tables. Show reasoning where there is room.*

**1.** Two parent functions pass through (1, 1) and (0, 0). Name them. __________________________

**2.** Which parent function never touches either axis? ______________________

**3.** Which two are reflections of each other across y = x? Give two different pairs.

____________________________________________

**4.** A graph has a minimum but no maximum, and no asymptotes. Name both possibilities.

__________________________________

**5.** f(x) = |x| on the interval [-3, 1]: minimum ____________ maximum ____________

**6.** Explain why a function can have a relative maximum on an interval but none overall.

__________________________________________________________

__________________________________________________________

**7.** Which parent function has two separate branches? ____________________

**8.** True or false: every parent function with an asymptote also has a restricted domain.

____________________________________________

**9.** Which two parent functions have both a minimum and pass through the origin? ______________________

## Exit Ticket

Two parent functions have domain and range that are both all real numbers, and both have 180° rotational symmetry.

Name them: ________________________ Name one function with a restricted range: ________________________

Name ______________________________________________ Date ___________ Period _____

# Modeling with a Parent Function: the Open Box

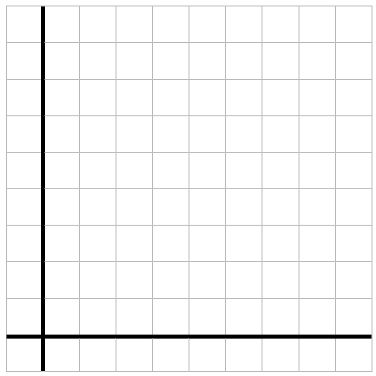
Algebra II | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

**Objective:** I can build a function that models a real situation, decide which domain makes sense for the context, and use a table or graph to find a maximum.

## The Problem

Start with a flat sheet of cardboard **20 inches by 10 inches**. Cut a square of side **x** out of each of the four corners, then fold the flaps up to make an open-top box.

*Sketch the sheet with the corner squares marked, and label one side x.*



## Part 1 — Build the Function

*Each cut removes x from BOTH ends of a side, so each side loses 2x.*

Height of the box: ____________________

Length of the box: ____________________ Width of the box: ____________________

Volume V(x) = length · width · height = ______________________________

Multiplied out: V(x) = ______________________________ Which parent function family? ______________

## Part 2 — Find the Domain That Makes Sense

*The equation accepts any real number. The cardboard does not.*

Why must x be greater than 0? ____________________________________________

The short side is 10 inches. What happens if x = 5? __________________________________________

So the contextual domain is ____________________ interval ________________ Open or closed? ________________________

## Part 3 — Make a Table

*Compute $V(x) = x(20 - 2x)(10 - 2x)$ for each value. Round to the nearest tenth.*

| x | 0.5 | 1 | 1.5 | 2 | 2.5 | 3 | 4 | 4.5 |
|---|---|---|---|---|---|---|---|---|
| V(x) | | | | | | | | |

Between which two x-values does the volume stop growing and start shrinking? ________________________

Largest volume in your table: ________________________ Is that certainly the true maximum? __________________________

## Part 4 — Refine the Maximum

*Test values between 2 and 2.5 to close in.*

| x | 2.0 | 2.1 | 2.11 | 2.2 | 2.3 |
|---|---|---|---|---|---|
| V(x) | | | | | |

Maximum volume ~ ____________________ when x ~ ________________

Box dimensions then: ______________________________

Range of the function on this domain: __________________________ interval __________________

## Part 5 — Interpret in Context

1. What does V(1) = 144 mean in words? __________________________________
2. Why is x = 5 excluded but x = 4.5 allowed? __________________________________
3. Would a domain of all reals make sense here? __________________________________

## Practice — A Second Box

*A sheet is 16 in. by 8 in. Squares of side x are cut from the corners. Show your work.*

**1.** Write the volume function.

__________________________________________________________

__________________________________________________________

V(x) = ______________________________

**2.** Which side limits the domain, and what is the limit? ____________________________

**3.** Contextual domain: __________________________

**4.** Compute V(1) and V(2). Show your arithmetic.

__________________________________________________________

__________________________________________________________

V(1) = __________________ V(2) = __________________

**5.** Explain why V(4) = 0 even though 4 is a positive number.

__________________________________________________________

__________________________________________________________

## Exit Ticket

A sheet is 12 in. by 12 in. Squares of side x are cut from the corners and the sides folded up.

Volume function: __________________________ Contextual domain: ________________________

Explain why the upper limit is 6: ________________________________________

Name ______________________________________________ Date ___________ Period _____

# Inverses from Graphs and Tables

Algebra II | Unit 1 | Day 10 | TEK A2.2.B

**Objective:** I can find the inverse of a relation from a table or a graph, use $f^{-1}(x)$ notation, and recognize that a function and its inverse are reflections across the line $y = x$.

## Vocabulary

**Inverse of a function** — the function that _____________ the original, written $f^{-1}(x)$. Swap x and y in a table; reflect across $y = x$ on a graph.

## Part 1 — Inverses from Ordered Pairs

*Swap the coordinates in each pair.*

**Example 1.** $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$

$f^{-1} = \{$ ________________________________________________ $\}$

Domain of f = __________________________ which becomes the ___________ of $f^{-1}$.

**You Try 1.** $g = \{ (1, 5), (2, 8), (3, 11) \}$  $g^{-1} = \{$ _________________________________ $\}$

## Part 2 — Inverses from a Table

*Fill in the inverse table by trading the rows.*

| **x** | -4 | 0 | 2 | 3 | 7 |
|---|---|---|---|---|---|
| **f(x)** | -6 | -1 | 4 | -7 | 7 |

*Now the inverse:*

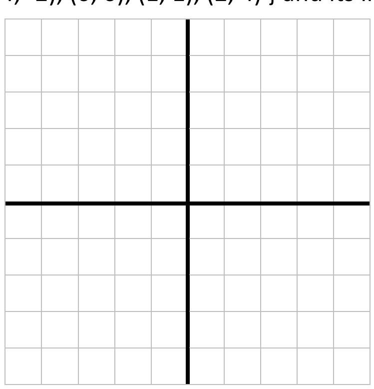
| **x** | -6 | -1 | 4 | -7 | 7 |
|---|---|---|---|---|---|
| **$f^{-1}(x)$** | | | | | |

The two tables are the same information, just _______________________.

## Part 3 — Graph a Function and Its Inverse

*Each square is 1 unit. Plot f, then plot $f^{-1}$, then draw the line $y = x$ through the origin at 45°.*

**A.** Plot $f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$ and its inverse.



$f^{-1} = \{$ ___________________________ $\}$

**B.** What do you notice?

The two graphs are mirror images across the line

_________________

Which points stay put under the swap?

___________________________

If (a, b) is on f, then on $f^{-1}$ you find

_________________

## Part 4 — Domain and Range Trade Places

For any relation: domain of f = _______________ of $f^{-1}$, and range of f = _______________ of $f^{-1}$.

| Function | Domain of f | Range of f | Domain of $f^{-1}$ | Range of $f^{-1}$ |
|---|---|---|---|---|
| $f(x) = x^3$ | all reals | all reals | ____________ | ____________ |
| $f(x) = 2^x$ | all reals | $(0, \infty)$ | ____________ | ____________ |
| $f(x) = \sqrt{x}$ | $[0, \infty)$ | $[0, \infty)$ | ____________ | ____________ |

Row 2 explains Day 7: the exponential and the _______________ are inverses, which is why their domains and ranges swapped.

## Part 5 — Quick Check

1. Does $f^{-1}(x)$ mean 1 divided by f(x)? _________________
2. If (3, -5) is on the graph of f, name a point on $f^{-1}$. ___________________
3. A graph and its inverse are reflections across which line? ___________________

## Practice

*Find each inverse. Show your work where there is room.*

**1.** p = { (0, 3), (1, 5), (2, 7) }  $p^{-1}$ = { ______________________________ }

**2.** Domain of p = _____________________  Range of $p^{-1}$ = _____________________

**3.** A table gives f: (2, 6), (4, 6), (5, 1). Write $f^{-1}$, then decide whether it is a function.

______________________________________________________________

______________________________________________________________

$f^{-1}$ = { ____________________________ }  A function? _________________________

**4.** That is a preview of Day 12. What went wrong in item 3?

______________________________________________________________

______________________________________________________________

**5.** If a graph contains (-3, 7), name the matching point on its inverse. _____________________

## Exit Ticket

A function h has the table below.

h: (1, 4), (2, 9), (5, 0)  $h^{-1}$ = { ________________________________ }

Domain of $h^{-1}$: _______________________  Range of $h^{-1}$: _______________________

Name ______________________________________________ Date ___________ Period _____

# Writing the Inverse of a Linear Function

Algebra II | Unit 1 | Day 11 | TEK A2.2.B

**Objective:** I can write the inverse of a linear function algebraically by swapping x and y and solving for y, then check my answer by graphing or with a table.

## Warm-Up — You Already Know How to Do This

*In Algebra I you solved literal equations for a variable. That is the whole skill today.*

Solve $y = 3x + 6$ for x: $x =$ ____________________________

## The Three Steps

**Step 1.** Replace f(x) with ________.

**Step 2.** _____________________.

**Step 3.** Solve for y, then write it as _______________.

## Part 1 — Worked Examples

**Example 1.** $f(x) = 2x + 3$

Swap: __________________________ Solve: __________________________

So $f^{-1}(x) =$ __________________________

**Example 2.** $f(x) = 3x - 5$

Swap: __________________________ Solve: __________________________

So $f^{-1}(x) =$ __________________________

## Part 2 — You Try

1. $f(x) = x + 7$

   $f^{-1}(x) =$ _______________________________

2. $f(x) = 4x$

   $f^{-1}(x) =$ _______________________________

3. $f(x) = 5x - 10$

   $f^{-1}(x) =$ _______________________________

4. $f(x) = -2x + 8$

   $f^{-1}(x) =$ _______________________________

## Part 3 — Check It Two Ways

Use $f(x) = 2x + 3$ and $f^{-1}(x) = (x - 3)/2$ from Example 1.

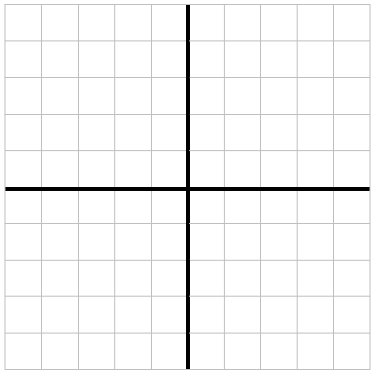
*Check 1: the tables should trade columns.*

| **x** | -1 | 0 | 1 | 2 |
|---|---|---|---|---|
| **f(x)** | 1 | 3 | 5 | 7 |

| **x** | 1 | 3 | 5 | 7 |
|---|---|---|---|---|
| **$f^{-1}(x)$** | | | | |

*Check 2: plot both. They must mirror across y = x.*

**Graph f and $f^{-1}$, plus the line y = x.**



**Record it.**

Slope of f: ____________

Slope of $f^{-1}$: ____________

What is the relationship?

__________________________

y-intercept of f is 3. Of $f^{-1}$?

________________

## Part 4 — Watch Out

A student says the inverse of f(x) = x + 4 is $f^{-1}(x) = 1/(x + 4)$. What went wrong?

___________________________________________________________

Which linear function is its own inverse? _______________________________

What is the inverse of a horizontal line, like f(x) = 3? ____________________________________

## Part 5 — In Context

To convert Celsius to Fahrenheit: F(c) = (9/5)c + 32.

Find the inverse: $F^{-1}(x)$ = ___________________________

What does the inverse do in words? __________________________________________

## Practice

*Write each inverse. Show the swap-and-solve steps.*

**1.** f(x) = (1/2)x - 6

_____________________________________________________________

_____________________________________________________________

$f^{-1}(x)$ = __________________________

**2.** f(x) = -x + 9

_____________________________________________________________

_____________________________________________________________

$f^{-1}(x)$ = __________________________

## Exit Ticket

Write the inverse of f(x) = 6x - 12, then verify it with one input.

$f^{-1}(x)$ = ___________________________

Check: f(3) = ____________ and $f^{-1}$(_________) = ____________

Name ____________________________________________ Date ___________ Period _____

# When Is an Inverse a Function?

Algebra II | Unit 1 | Day 12 | TEK A2.2.C

**Objective:** I can decide whether the inverse of a function is itself a function, and restrict the domain of the original when it is not.

## Part 1 — The Problem with $f(x) = x^2$

*Fill in the table, then look for repeats in the bottom row.*

| **x** | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| **$f(x) = x^2$** | | | | | | | |

Which outputs appear more than once? ___________________________ So f is ________ one-to-one.

Now swap to build the inverse. Both (1, 1) and (1, -1) appear. Is that a function? ____________

Why not? ______________________________________________

Written out, the inverse relation is $y =$ ________________, and the ± is exactly the problem.

## Part 2 — Fix It with a Restriction

*Keep only half of the parabola, so no output repeats.*

| **Restriction on $f(x) = x^2$** | **Which half of the parabola** | **Then $f^{-1}(x) =$** |
|---|---|---|
| $x \geq 0$ | _________________ | _____________ |
| $x \leq 0$ | _________________ | _____________ |

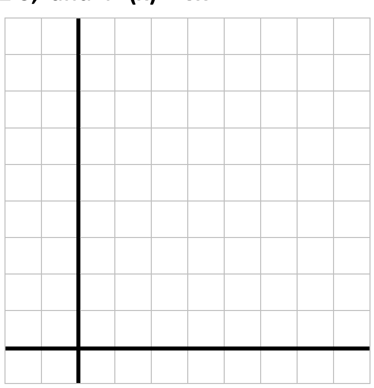
Either restriction works. The usual choice is ______________, which gives the familiar $f^{-1}(x) = \sqrt{x}$.

Restricting the domain of f also restricts the ______________ of $f^{-1}$.

## Part 3 — Graph the Restricted Pair

*Each square is 1 unit. Graph only the half of $x^2$ you kept, then its inverse, plus $y = x$.*

**$f(x) = x^2$ for $x \geq 0$, and $f^{-1}(x) = \sqrt{x}$**



**Record it.**

Domain of restricted f: ______________

Range of restricted f: ______________

Domain of $f^{-1}$: ______________

Do they mirror across $y = x$? ____________

## Part 4 — Which Pairs Need No Restriction?

*Some parent functions are already one-to-one, so their inverses are functions with no help.*

| Function | One-to-one? | Restriction needed? | Its inverse |
|---|---|---|---|
| $f(x) = x^2$ | ___________ | _____________ | _____________ |
| $f(x) = x^3$ | ___________ | _____________ | _____________ |
| $f(x) = 2^x$ | ___________ | _____________ | _____________ |
| $f(x) = 2x + 1$ | ___________ | _____________ | _____________ |
| $f(x) = \|x\|$ | ___________ | _____________ | _____________ |

Pattern: functions that only _____________ (or only decrease) are one-to-one. Ones that turn around, like $x^2$ and $|x|$, are not.

## Part 5 — Quick Check

1. Is the inverse of every function also a function? ________________
2. Which test tells you the inverse will be a function? ________________________
3. Why does $x^3$ need no restriction but $x^2$ does? ________________________________

## Practice

*Decide whether a restriction is needed, and give one if so.*

**1.** Does $f(x) = |x|$ have an inverse that is a function with no restriction? ______________

**2.** Restrict $f(x) = |x|$ so its inverse is a function. Give the restriction and the inverse.

________________________________________

**3.** Is $f(x) = x$ to the 4th power one-to-one? ____________ Give a restriction that fixes it. ________________

**4.** True or false: $f(x) = x^3 + 1$ needs a domain restriction for its inverse to be a function.

________________________________________

**5.** Explain, using the horizontal line test, why $f(x) = x^2 - 4$ has an inverse that is not a function.

__________________________________________________________

__________________________________________________________

**6.** Give one restriction (there are two possible) that makes $f(x) = x^2 - 4$ a one-to-one function.

________________________

## Exit Ticket

Consider $f(x) = x^2 - 4$ with no restriction.

Is its inverse a function? ______________ Give a restriction that fixes it: ________________________

Explain in one sentence why the restriction is needed: ____________________________________________

Name ____________________________________________ Date ___________ Period _____

# Composition to Verify Inverses

Algebra II | Unit 1 | Day 13 | TEK A2.2.D

**Objective:** I can compose two functions and use the results $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ to prove whether they are inverses of each other.

## Vocabulary

**Composition of functions** — putting one function ____________ another and evaluating: f(g(x)), read "f of g of x." Work from the ______________.

## Part 1 — Warm Up on Composition

Let f(x) = 3x - 5 and g(x) = (x + 5)/3. Work inside first.

g(2) = ______________ f(4) = ______________ f(g(4)) = f(________) = ____________

g(f(4)) = g(________) = ____________ Notice both came back to ________, the number we started with.

## Part 2 — Prove It with Algebra

**Example 1.** f(x) = 3x - 5 and g(x) = (x + 5)/3

f(g(x)) = 3 · ____________________ - 5 = __________________ = ________

g(f(x)) = (________________ + 5)/3 = ________________ = ________

Both give x, so f and g __________ inverses.

**You Try 1.** f(x) = 2x + 3 and g(x) = (x - 3)/2

f(g(x)) = ________________________ g(f(x)) = ________________________ Inverses? ____________

**You Try 2.** f(x) = x + 6 and g(x) = x - 6

f(g(x)) = ________________________ g(f(x)) = ________________________ Inverses? ____________

## Part 3 — A Pair That Is NOT Inverse

Test f(x) = 2x + 1 and g(x) = 2x - 1. Do not assume - compute.

f(g(x)) = 2(________________) + 1 = __________________ Does this equal x? __________

So they are __________ inverses. What is the actual inverse of f(x) = 2x + 1? $f^{-1}(x)$ = ____________________

Lesson: changing a sign is __________ the same as undoing the function. You must undo every step, in reverse order.

## Part 4 — Error Analysis

A student is given $f(x) = 2(x - 3)^2 + 4$ and claims $f^{-1}(x) = \sqrt{(x - 3)} + 4$.

Test it: is $f(f^{-1}(x)) = x$? ____________ So the claim is ____________.

Find the mistake. Undo the steps in reverse: subtract 4, divide by 2, take the square root, then add 3.

Correct inverse: $f^{-1}(x)$ = ____________________________

Why does a ± show up here? ____________________________________________

What restriction would make the inverse a function? __________________________

## Part 5 — Self-Inverse Functions

*A few functions are their own inverse - composing them with themselves gives x.*

1. $f(x) = x$ $f(f(x)) =$ ________________ Self-inverse? ____________
2. $f(x) = -x$ $f(f(x)) =$ ________________ Self-inverse? ____________
3. $f(x) = 1/x$ $f(f(x)) =$ ________________ Self-inverse? ____________

What do their graphs share? ______________________________________

## Practice

*Decide whether each pair is inverse. Show BOTH compositions - one direction is not proof.*

**1.** $f(x) = 5x$ and $g(x) = x/5$

___________________________________________________________

___________________________________________________________

$f(g(x)) =$ ________________ $g(f(x)) =$ ________________ Inverses? ____________

**2.** $f(x) = x - 3$ and $g(x) = 3 - x$

___________________________________________________________

___________________________________________________________

$f(g(x)) =$ ________________ Inverses? ____________ Correct inverse of f: __________________

**3.** $f(x) = (x + 1)/2$ and $g(x) = 2x - 1$

___________________________________________________________

___________________________________________________________

$f(g(x)) =$ ________________ $g(f(x)) =$ ________________ Inverses? ____________

**4.** $f(x) = x^2$ and $g(x) = \sqrt{x}$. Test with $x = -4$ as well as $x = 4$.

___________________________________________________________

___________________________________________________________

Does it work for every real x? __________________________ Restriction needed: ________________

**5.** Write the inverse of $f(x) = 7x + 2$, then verify it by composition.

___________________________________________________________

___________________________________________________________

___________________________________________________________

$f^{-1}(x) =$ ________________________

## Exit Ticket

Decide whether $f(x) = 4x - 8$ and $g(x) = (x + 8)/4$ are inverses. Show both compositions.

$f(g(x)) =$ ______________________ $g(f(x)) =$ ______________________ Inverses? ____________

Why is checking only one direction not enough? ________________________________________